

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	B.Tech AI &DS	Date:	14-08-2026

Code of Conduct

I ____N.Kishan Kumar____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____N.Kishan Kumar_____

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Reference Resolution Algorithm

Aim

To design an algorithm that identifies pronouns in a discourse and links them to their correct antecedents or entities using grammatical agreement and discourse context.

Given Discourse

"Ravi met Arun at the library. He borrowed a book and later returned it."

Algorithm

Input: A discourse containing entities and pronouns.

Output: Discourse with pronouns replaced by their corresponding entities.

Pseudocode

Algorithm ReferenceResolution(Discourse)

1. Split the discourse into individual sentences.
2. Identify all noun phrases and named entities.
3. Identify all referring expressions such as pronouns.
4. For each pronoun:
 - a. Find possible antecedents in the previous discourse.
 - b. Check gender and number agreement.
 - c. Check grammatical compatibility.
 - d. Apply recency preference.
 - e. Consider the semantic role of each candidate.
5. Select the most suitable antecedent.
6. Replace the pronoun with the selected antecedent.
7. Return the resolved discourse.

End Algorithm

Application to the Given Discourse

The entities introduced in the first sentence are Ravi and Arun.

The pronouns are:

- He – refers to Ravi.
- it – refers to the book.

The second sentence states, "He borrowed a book." Since Ravi is the preferred recent and contextually suitable antecedent, "He" is resolved as Ravi.

The pronoun "it" occurs after the noun phrase a book and agrees with the singular inanimate object. Therefore, "it" refers to the book.

Resolved Discourse

"Ravi met Arun at the library. Ravi borrowed a book and later returned the book."

Validation

The resolution is based on discourse context rather than simply selecting any preceding noun. The algorithm considers the entities introduced in the previous sentence, their grammatical properties, recency, and semantic compatibility. "He" is resolved to Ravi because Ravi is the most suitable antecedent in the context. "It" is resolved to the book because a book is the immediately preceding compatible noun phrase and can be returned. Thus, the resulting discourse is grammatically and semantically coherent.

2. Text Coherence and Discourse Structure Algorithm

Aim

To design an algorithm that identifies logical relationships between sentences, such as cause-effect, sequence, and elaboration, and constructs a coherent discourse structure.

Given Discourse

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Algorithm

Input: A sequence of sentences.

Output: Discourse relations and a discourse structure.

Pseudocode

Algorithm DiscourseStructureAnalysis(Text)

1. Split the text into sentences S1, S2, ..., Sn.
2. Identify keywords and discourse markers.
3. Determine the semantic relationship between consecutive sentences.
4. If a sentence explains the reason for another sentence:

Mark the relation as CAUSE.
5. If a sentence describes the result of a previous sentence:

Mark the relation as EFFECT.
6. If events occur one after another:

Mark the relation as SEQUENCE.
7. If a sentence provides additional information:

Mark the relation as ELABORATION.
8. Construct a discourse tree or relation graph.
9. Return the identified relations and structure.

End Algorithm

Application to the Given Discourse

There are three sentences:

- **S1:** The roads were flooded after heavy rainfall.
- **S2:** Therefore, schools were closed for the day.
- **S3:** Students attended classes online.

The word "**Therefore**" is an important discourse marker. It indicates that S2 is a consequence or effect of the situation described in S1.

The relationship between S1 and S2 is:

S1 → CAUSE → S2

The closure of schools leads to students attending classes online. Therefore, S2 provides the context for S3.

S2 → RESULT/CONSEQUENCE → S3

Discourse Structure Representation

Heavy rainfall



Roads were flooded (S1)



Schools were closed (S2)



Students attended classes online (S3)

Or:

S1: Roads were flooded



| Cause



S2: Schools were closed



| Consequence



S3: Students attended classes online

Validation

The discourse is coherent because each sentence is logically connected to the next. Heavy rainfall causes flooding, flooding results in school closure, and school closure leads to online classes. The cause-effect and consequence relationships allow the reader to understand the sequence of events easily. The discourse marker "Therefore" explicitly signals the relationship between the first and second sentences.

3. Dialogue Act Identification Algorithm

Aim

To design an algorithm that identifies the communicative function or dialogue act of each utterance in a conversation.

Given Conversation

User: "Can you book a train ticket for me?"

Agent: "Sure, where would you like to travel?"

User: "I want to go to Chennai."

Agent: "Your ticket has been booked."

Dialogue Acts

The possible dialogue acts are:

- Request
- Question
- Inform
- Confirmation
- Action

Pseudocode

Algorithm DialogueActIdentification(Conversation)

1. Read each utterance in the conversation.
2. Analyze keywords, sentence structure, and communicative intention.
3. If the user asks the agent to perform a task:
 Classify as REQUEST.
4. If the utterance asks for information:
 Classify as QUESTION.
5. If the user provides information:
 Classify as INFORM.
6. If the agent confirms that a requested task is completed:
 Classify as CONFIRMATION.
7. If the utterance represents the execution of a task:
 Classify as ACTION.
8. Store the dialogue acts in conversational order.
9. Return the dialogue-act sequence.

End Algorithm

Application

Speaker	Utterance	Dialogue Act	Reason
User	"Can you book a train ticket for me?"	Request	The user asks the agent to perform an action.
Agent	"Sure, where would you like to travel?"	Question	The agent asks for required information.
User	"I want to go to Chennai."	Inform	The user provides the destination.
Agent	"Your ticket has been booked."	Confirmation	The agent confirms completion of the requested task.

Dialogue-Act Sequence

Request → Question → Inform → Confirmation

Evaluation

Dialogue acts help the conversational agent understand the user's intention and determine the next appropriate response. The first utterance is recognized as a Request, so the agent knows that a booking task must be performed. The agent then asks a Question to obtain missing information. The user's Inform act provides the destination, and the final Confirmation indicates that the requested action has been completed.

Thus, dialogue-act identification supports effective conversational flow and allows the agent to maintain the correct context throughout the interaction.

4. Language Generation and Surface Realization Algorithm

Aim

To design an algorithm that converts a semantic representation into a grammatically correct natural language sentence using lexical selection, sentence structuring, and surface realization.

Semantic Input

Action: Buy

Agent: Student

Object: Book

Tense: Past

Algorithm

Input: Semantic representation containing action, agent, object, and tense.

Output: Grammatically correct natural language sentence.

Pseudocode

Algorithm SurfaceRealization(SemanticInput)

1. Extract the semantic roles:

Agent

Action

Object

Tense

2. Select appropriate lexical items:

Agent → student

Action → bought

Object → book

3. Apply the specified tense:

Convert "buy" into the past tense "bought".

4. Arrange the sentence using English SVO structure:

Subject + Verb + Object

5. Add required determiners:

"the student" and "a book".

6. Perform grammatical agreement and word-order checking.

7. Generate the final sentence.

End Algorithm

Processing Steps

Step 1 – Lexical Selection

The semantic action Buy is mapped to the lexical verb bought because the required tense is past.

Step 2 – Sentence Structuring

English normally follows the Subject–Verb–Object (SVO) structure.

Subject → The student

Verb → bought

Object → a book

Step 3 – Surface Realization

The components are combined to form the final sentence.

Final Generated Sentence

"The student bought a book."

Validation

The generated sentence is grammatically correct because it follows the English SVO structure. The subject "student" is singular and the verb "bought" is the correct past-tense form of "buy". The object "a book" is also grammatically appropriate. Therefore, the semantic representation has been successfully converted into a natural and grammatically correct sentence.

5. Hybrid Interlingua-Based and Statistical Machine Translation Algorithm

Aim

To design an algorithm that combines Interlingua-based Machine Translation with statistical translation techniques and apply it to the sentence:

"The boy is playing football."

Algorithm

Input: Source-language sentence.

Output: Translated target-language sentence.

Pseudocode

Algorithm HybridMachineTranslation(SourceSentence)

1. Analyze the source sentence.
2. Perform lexical and syntactic analysis.
3. Identify:

Subject

Action

Object

Tense

Aspect

4. Convert the analyzed meaning into an Interlingua representation.
 5. Generate multiple candidate translations in the target language.
 6. Calculate statistical scores for each candidate.
 7. Consider:
 - Translation probability
 - Language-model probability
 - Word order
 - Grammatical correctness
 8. Select the candidate with the highest overall score.
 9. Generate the final target-language sentence.
 10. Return the translation.
- End Algorithm

Step 1 – Source Sentence Analysis

The source sentence is:

"The boy is playing football."

Its components are:

- **Subject:** The boy
- **Action:** Playing
- **Object:** Football
- **Tense:** Present
- **Aspect:** Continuous

The sentence describes an ongoing action performed by a boy.

Step 2 – Interlingua Representation

The language-independent meaning can be represented as:

AGENT: BOY

ACTION: PLAY

OBJECT: FOOTBALL

TENSE: PRESENT

ASPECT: CONTINUOUS

A structured representation can also be written as:

PLAY(

AGENT = BOY,

OBJECT = FOOTBALL,

TENSE = PRESENT,

ASPECT = CONTINUOUS

)

This representation captures the meaning without depending on the grammar or word order of a particular language.

Step 3 – Candidate Target-Language Translations

Suppose the target language is **Telugu**. The system may generate candidate translations based on the Interlingua representation.

Possible candidates include:

1. బాలుడు ఫుట్ బాల్ ఆడుతున్నాడు.
2. అబ్బాయి ఫుట్ బాల్ ఆడుతున్నాడు.
3. బాలుడు ఫుట్ బాల్ ఆడాడు.

The third candidate represents a completed past action and therefore does not correctly preserve the continuous present meaning.

Step 4 – Statistical Scoring

The statistical translation component evaluates the candidates using translation probability and language-model probability.

A simplified scoring function can be represented as:

Score(Translation)

=

Translation Probability

+

Language Model Probability

+

Grammar Score

+

Semantic Compatibility

The candidate "బాలుడు ఫుట్ బాల్ ఆడుతున్నాడు." receives a high score because it correctly represents the subject, action, object, and ongoing present action.

Step 5 – Final Translation

Final Translation:

"బాలుడు ఫుట్ బాల్ ఆడుతున్నాడు."

English meaning: "The boy is playing football."

Evaluation

The Interlingua representation improves translation because it separates the underlying meaning from the grammatical structure of the source language. This helps the system preserve important semantic information such as the agent, action, object, tense, and aspect.

Statistical methods improve the process by comparing possible target-language translations and selecting the candidate that is most probable and grammatically natural. Therefore, combining Interlingua-based representation with statistical scoring reduces ambiguity, improves grammatical accuracy, and produces more natural translations.

Overall Conclusion

The five algorithms demonstrate important Natural Language Processing techniques. Reference resolution identifies the correct antecedents of pronouns, discourse analysis establishes relationships between sentences, dialogue-act identification determines conversational intentions, surface realization converts semantic structures into grammatical sentences, and hybrid machine translation combines language-independent meaning with statistical selection. These techniques enable NLP systems to understand, interpret, and generate human language more effectively.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____